



PCT62 – Temperature Control

The Armfield PCT62 The system allows users to adjust the heater power and the air flow rate to develop a PID based control system then adjust these parameters to achieve the required time/temperature change profile for the system in response to step changes in system requirements.

PRODUCT CODE: PCT62

Categories: Educational, PCT Series - Process Control Teaching System, PCT62 - Temperature Control

Tags: Air Temperature , Armfield , ArmSOFT , Data Logging , Direct Heat , External Controller , Indirect Heat , PCT62 , PID Controller , Process Control Technology , Sensor Calibration , Surface Temperature , Temperature Control

DESCRIPTION

Temperature control is a critical aspect of many industries, including chemical, pharmaceutical, food and beverage, manufacturing, and HVAC (heating, ventilation, and air conditioning). Understanding temperature control is crucial for students pursuing careers in these industries.

Proper temperature control can help industries reduce energy consumption and operating costs. Students learn how to optimize temperature control systems to minimize energy waste and environmental impact.

The Armfield PCT62 temperature control system introduces students to fundamental control concepts, such as feedback control, proportional-integral-derivative (PID) controllers, and closed-loop control systems.

The temperature process control system includes a heated plate within a duct and a thermocouple. A fan at one end of the duct blows ambient air over the block, to change the control conditions and provide a disturbance to the system.

The system allows users to adjust the heater power and the air flow rate to develop a PID based control system then adjust these parameters to achieve the required time/temperature change profile for the system in response to step changes in system requirements.

TECHNICAL SPECIFICATIONS

PCT62 TEMPERATURE CONTROL PROCESS

A temperature control process trainer, comprising:

- 4mm x 1.5mm Wall clear PVC tube (300mm)
- 24V PSU 60w 2.5a
- Flow process control assembly
- Barley pot for process control
- USB lead
- Manual Control software allowing low level access to the Drive and Load in each system allowing calibration of sensors and drive systems
- On/Off Control software allowing control of each system with a simple On/Off algorithm, view software based oscillations and to explore the effects of hysteresis
- PID Control software allowing users to enter values for Kp, Ki, Kd and see how the system reacts to inputted values

FEATURES & BENEFITS

- USB, Wi-Fi, Bluetooth and LAN communications supplied as standard
- Supplied software includes Basic control, On/Off control and
- PID control
- Software dynamically displays Set point, Process value and Kp, Ki, and Kd
- MATLAB and LabVIEW compatible
- Supplied with full set of manuals and teaching material

Demonstration Capabilities

- Understanding how to control driving devices
- Understanding the sensors
- On/Off control systems
- System time constant
- P controller
- PI controller
- PID controller
- Zeigler Nichols algorithm
- Integral wind up
- Derivative filter
- Manual tuning
- Interfacing with MATLAB/LabVIEW

PCT62 Video



Video del PCT62

Requirements

Mains electrical supply: 110-230V, AC 50-60 Hz.

PC and Display meeting the following minimum specification:

- **Processor:** 1Ghz or faster
- **RAM:** 1Gb or more
- **HDD Space:** 1Gb
- **OS:** 32 or 64bit Windows 7, 8, 10 or 11
- **Display:** Recommended minimum (1920 by 1080) full HD

Shipping Specifications



PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.095m³

Gross Weight: 12.1Kg

Dimensions

Length: 0.64m

Width: 0.45m

Height: 0.33m

Order Codes

- **PCT62-UK:** Temperature Control System
- **PCT62-EU:** Temperature Control System
- **PCT62-USA:** Temperature Control System

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